

Spring 2024

Problem 1 (Sp24). Differentiate $f(x) = \begin{cases} x^2 + 1 & \text{if } x > 1 \\ 2x & \text{if } x \leq 1 \end{cases}$ and carefully explain, using the definition of the derivative, what happens at $x = 1$.

Problem 2 (Sp81, Sp24). 1. Give an example of a sequence of $\mathcal{C}^1$ functions

$$f_k : [0, \infty) \rightarrow \mathbb{R} \quad k = 1, 2, 3, \dots$$

such that $f_k(0) = 0$ for all k , and $f'_k(x) \rightarrow f'(x)$ for all x as $k \rightarrow \infty$, but $f_k(x)$ does not converge to $f(x)$ for all x as $k \rightarrow \infty$.

2. State an extra condition which would imply that $f_k(x) \rightarrow f(x)$ for all x as $k \rightarrow \infty$.

Problem 3 (Sp86, Sp24). Let f be a real valued continuous function on $\mathbb{R}$ satisfying the *mean value inequality* below:

$$f(x) \leq \frac{1}{2h} \int_{x-h}^{x+h} f(y) \, dy \quad x \in \mathbb{R} \quad h > 0$$

Prove:

1. The maximum of f on any closed interval is assumed at one of the endpoints.
2. f is convex.

Problem 4 (Fa90, Sp24). Let the function f be analytic in the entire complex plane and satisfy

$$\int_0^{2\pi} |f(re^{i\theta})| \, d\theta \leq r^{17/3}$$

for all $r > 0$. Prove that f is the zero function.

Problem 5 (Sp24). Let f be the meromorphic function on $\mathbb{C}$ defined by

$$f(z) = 1 + \frac{1}{z} + \frac{1}{z^2}$$

Use the argument principle to show that

$$\int_C \frac{f'(z)}{f(z)} \, dz = 0$$

for all positively oriented simple closed contours C in $U = \{z \in \mathbb{C} \mid |z| > 1\}$.

Problem 6 (Fa81, Sp24). Let $A, B \in M_{2 \times 2}$ be given by:

$$A = \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix} \quad B = \begin{pmatrix} 2 & 1 \\ 0 & 4 \end{pmatrix}$$

Define a linear transformation $L : M_{2 \times 2} \rightarrow M_{2 \times 2}$ by $L(X) = AXB$. Compute the trace and the determinant of L .

Problem 7 (Fa93, Sp24). Let w be a positive continuous function on $[0, 1]$, n a positive integer, and P_n the vector space of real polynomials whose degrees are at most n , equipped with the inner product

$$\langle p, q \rangle = \int_0^1 p(t)q(t)w(t) dt$$

1. Prove that P_n has an orthonormal basis $p_0, p_1, \dots, p_n$, i.e., $\langle p_j, p_k \rangle = 1$ for $j = k$ and 0 for $j \neq k$ such that $\deg p_k = k$ for each k .
2. Prove that $\langle p_k, p'_k \rangle = 0$ for each k .

Problem 8 (Fa94, Fa13, Sp24). Let R be a ring with identity, and let u be an element of R with a right inverse. Prove that the following conditions on u are equivalent:

1. u has more than one right inverse;
2. u is a zero divisor;
3. u is not a unit.

Problem 9 (Sp79, Sp97, Sp24). Classify all abelian groups of order 80 up to isomorphism.

Problem 10 (Sp24). Give an epsilon–delta proof that $\lim_{x \rightarrow 5} x^2 = 25$.

Problem 11 (Sp24). Let (X, d) be a metric space, and let $\rho: X \times X \rightarrow \mathbb{R}$ be the function defined by

$$\rho(x, y) = \frac{d(x, y)}{1 + d(x, y)}$$

1. Show that ρ is a metric on X .
2. Show that the metrics d and ρ are equivalent.

Problem 12 (Fa99, Sp24). Let f be a continuous real valued function on $[0, \infty)$ such that $\lim_{x \rightarrow \infty} f(x)$ exists (finitely). Prove that f is uniformly continuous.

Problem 13 (Su77, Sp24). Let f be an analytic function such that

$$f(z) = 1 + 2z + 3z^2 + \cdots \quad \text{for } |z| < 1$$

Define a sequence of real numbers $a_0, a_1, a_2, \dots$ by

$$f(z) = \sum_{n=0}^{\infty} a_n (z + 2)^n$$

Find the radius of convergence of the series

$$\sum_{n=0}^{\infty} a_n z^n$$

Problem 14 (Fa97, Fa10, Sp24). Evaluate the integral $\int_{-\infty}^{\infty} \frac{\cos kx}{1 + x + x^2} dx$, where $k \geq 0$.

Problem 15 (Fa94, Sp24). Prove that the matrix

$$\begin{pmatrix} 1 & 1.00001 & 1 \\ 1.00001 & 1 & 1.00001 \\ 1 & 1.00001 & 1 \end{pmatrix}$$

has one positive and one negative eigenvalue.

Problem 16 (Sp77, Su82, Sp24). A square matrix A is nilpotent if $A^k = 0$ for some positive integer k .

1. If A and B are nilpotent, is $A + B$ nilpotent?
2. Prove: If A and B are nilpotent matrices and $AB = BA$, then $A + B$ is nilpotent.
3. Prove: If A is nilpotent then $I + A$ and $I - A$ are invertible.

Problem 17 (Fa83, Sp24, Sp26). 1. Let R be a finite integral domain, i.e., a finite commutative ring with 1 and no zero divisors. Prove that R is a field.

2. Show that if p is the characteristic of R then the Frobenius map $F : R \rightarrow R$, $F(x) = x^p$ is an automorphism (and in particular is bijective).

Problem 18 (Sp24). If q is a rational number, the *period* of q in base n is the least number a such that the base n representation of q eventually consists of a repeating sequence of length a . For example, the period of $1/7$ in base 10 is 6 since $1/7 = 0.\overline{142857}$.

Suppose p is a prime number. Show that for any $n > 1$ such that $p \nmid n$, the period of $1/p$ in base n is at most $p - 1$, and this maximum is achieved for some n .